

Measure of dispersion:

Data set-1
10, 10, 5, 5, 30, 0
mean = 10

$$\text{range} = 30 - 0 = 30$$

Data set-2
8, 9, 10, 11, 12.

$$\text{mean} = 10$$

$$\text{range} = 12 - 8 = 4$$

Data set-1
10, 11, 11.5, 11.6, 20
range = 10

Data set-2.
10, 12, 15, 18, 20
range = 10

$$\text{Variance} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\text{Standard deviation} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\text{Mean absolute deviation about a point A} = \frac{1}{n} \sum_{i=1}^n |x_i - A|$$

$$\frac{1}{n} \sum_{i=1}^n (x_i - A)^2 \geq \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\frac{1}{n} \sum_{i=1}^n |x_i - A| \geq \frac{1}{n} \sum_{i=1}^n |x_i - \text{median}|$$

$$\frac{1}{n} \sum (x_i - A)^2 \text{ — mean square deviation about a point A.}$$

Value	freq
x_1	f_1
x_2	f_2
$\vdots$	$\vdots$
x_k	f_k
	n

$$\bar{x} = \frac{1}{n} \sum_{i=1}^k x_i f_i$$

$$\text{Variance} = \frac{1}{n} \sum_{i=1}^k (x_i - \bar{x})^2 f_i$$

for group frequency distribution for continuous variable
 $x_i \rightarrow$ class mark of the i th class.

$$x_1, \dots, x_n \rightarrow \frac{n^2}{n}$$

$$a + bx_1, a + bx_2, \dots, a + bx_n \rightarrow \text{mean } a + b\bar{x}$$

$$\frac{1}{n} \sum_{i=1}^n (a + bx_i - a - b\bar{x})^2$$

$$= b^2 \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$= b^2 \sigma^2$$

coefficient of variation (C.V.)

Coefficient of variation	
Data set 1	Data set - 2
10, 11, 12, 13	1001, 1002, 1003, 1004.
Var = 1.25	Var = 1.25.
mean = 11.5	mean = 1002.5.
CV = 9.7%	CV = 1.1%
CV $CV = \frac{sd}{mean} \times 100$	

- Data sets have different units and/or measurement.
- Data sets have different means.

Moments

$$m_r = \frac{1}{n} \sum_{i=1}^n (x_i - A)^r \quad r = 0, 1, 2, 3, \dots$$

$\rightarrow$ r th moment about the point A.

A = 0, $m_r = \frac{1}{n} \sum_{i=1}^n x_i^r$ r th raw moment.

A = $\bar{x}$ $m_r = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^r$ r th central moment.

$$m_0 = 1, m_1 = 0$$

$$m'_0 = 1$$

$$m_1 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})$$

$$= \frac{1}{n} \sum_{i=1}^n x_i - \frac{1}{n} \sum_{i=1}^n \bar{x}$$

$$= \bar{x} - \bar{x}$$

$$= 0$$

$$\frac{1}{n} \sum_{i=1}^n \bar{x}$$

$$= \bar{x} \frac{1}{n} \sum_{i=1}^n 1$$

$$= \bar{x} \times \frac{n}{n}$$

$$m_2 = m'_2 - (m'_1)^2$$

$$m_3 = m'_3 - 3m'_2 m'_1 + 2(m'_1)^3$$

$$m_4 = m'_4 - 4m'_3 m'_1 + 6m'_2 (m'_1)^2 - 3(m'_1)^4$$

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^r = \frac{1}{n} \sum_{i=1}^n (x_i - A + A - \bar{x})^r$$

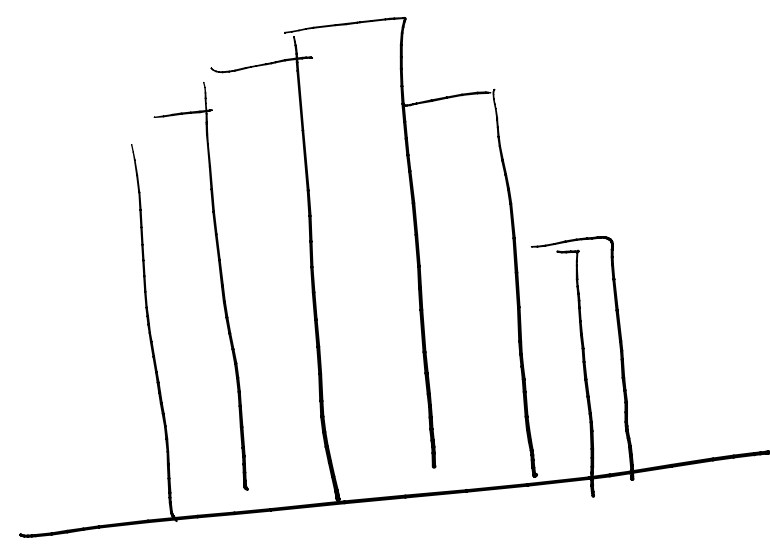
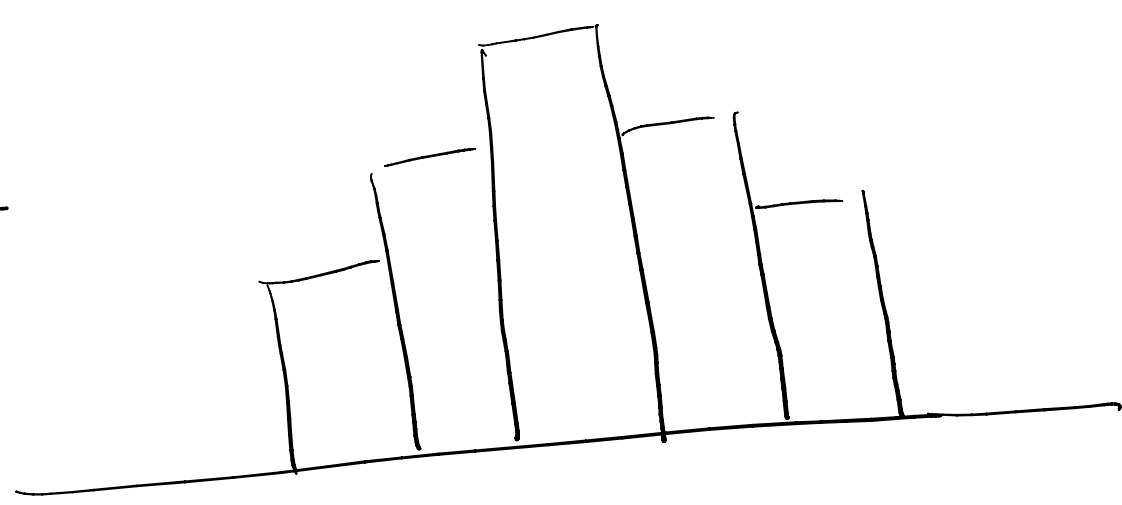
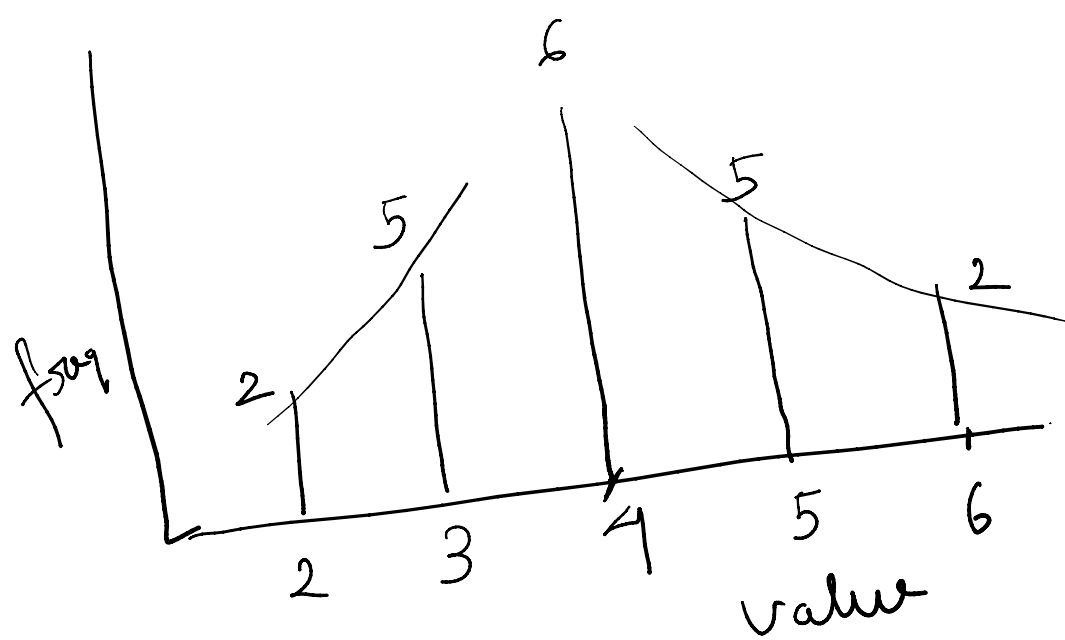
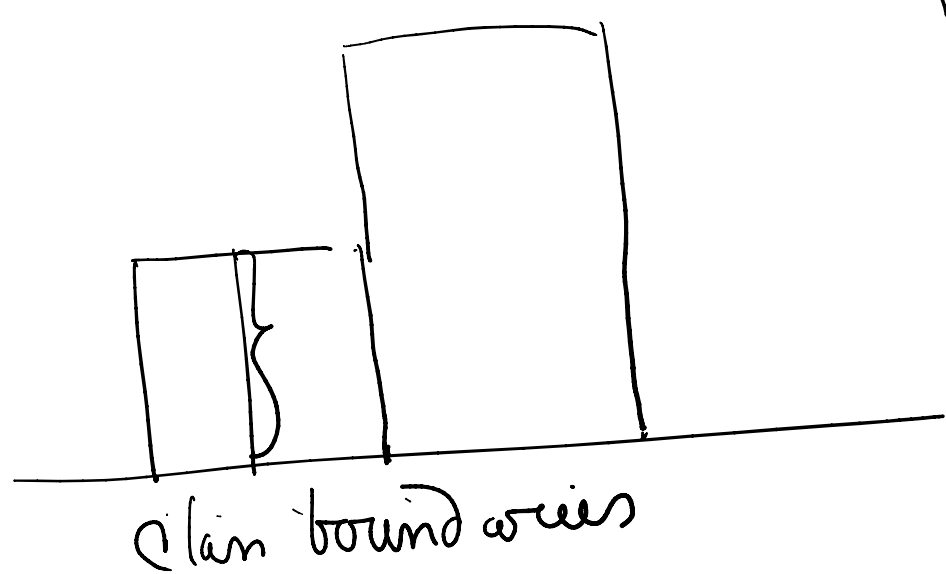
$$(\bar{x} - A) = \frac{1}{n} \sum_{i=1}^n x_i - \frac{n}{n} A$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i - A) = m'_1$$

$$m_r = \text{about } \bar{x}$$

$$m'_r = \text{about } A$$

Skewness:



Skewness — of a frequency distribution, the mean the degree of its departure from symmetry.

For discrete variable, frequency distribution is symmetrical about the value x_0 , if the frequency of $x_0 + h$ is same as frequency of $x_0 - h$ whatever h may be.

In case of cont variable, ~~it is so~~ the frequency distribution is symmetrical about a point x_0 , if the frequency density at $x_0 + h$ is same as the frequency density at $x_0 - h$.